JAIN Knowledge Campus, Jayanagar 9th Block, Bengaluru

DEPARTMENT OF COMPUTER SCIENCE AND IT

MCA PROGRAMME

ACADEMIC YEAR: 2026-2027

EXP: 10 - OPTIMIZE A MACHINE LEARNING MODEL BY TUNING HYPERPARAMETERS USING AZURE ML HYPERDRIVE

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SEM: III SPEC: SCT

SUBJECT: CLOUD ANALYTICS-LAB

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1 Aim

To optimize a machine learning model by tuning its hyperparameters in Azure Machine Learning (Azure ML), using the student performance dataset from Experiments 07 to 09, and to select the best configuration by a measured error metric, not by guesswork.

2 Objective and Theory

The objectives are to create an Azure ML workspace and a compute cluster, register the dataset as a data asset, define a hyperparameter search space for a Random Forest regressor, run a sweep job that trains several configurations, compare the trials by root mean squared error (RMSE), and register the best model for deployment in Experiment 11.

Hyperparameter tuning is the search over model settings that are fixed before training, such as the number of trees in a forest, the maximum tree depth, and the minimum samples required to split a node [1]. HyperDrive is the Azure ML name for this sweep process in the version 1 software development kit (SDK), and in the current Azure ML command line interface (CLI) version 2 the same process is a sweep job [2]. This experiment uses a CLI version 2 sweep job, and the report uses the experiment title HyperDrive for that sweep, because the search, the primary metric, and the best trial selection are the HyperDrive process, even though the job type shown in Azure ML Studio is Sweep.

The model is a RandomForestRegressor that predicts MathScore from six features: Age, Gender encoded as Male 1 and Female 0, ReadingScore, WritingScore, AttendanceRate, and StudyHoursPerWeek. The data are split into 80 percent training and 20 percent test, with random_state 42, giving 40 training rows and 10 test rows from the 50 rows. The primary metric is RMSE, with the goal minimize, and R2 is recorded as a secondary metric. RMSE is in the same units as MathScore, so a lower RMSE means predictions closer to the actual scores on the test set.

Missing values are handled by median imputation, not by dropping rows. In the 50 row file, Age, AttendanceRate, and StudyHoursPerWeek each have 5 missing values, in different rows, so dropping all rows with any missing feature would leave only 35 rows. Median imputation replaces each missing value with the median of that column, so all 50 rows are used. This is stated here so that the results in Section 5 can be reproduced exactly from train.py in the GitHub folder.

3 Architecture Flow

Data flows from Blob Storage into the Azure ML workspace as a data asset, the sweep job trains 8 Random Forest configurations on the compute cluster, and the best configuration is retrained and registered as a model, as shown in Figure 1.1. Table 1.1 lists the resources used.

Table 1.1 Resources used in Experiment 10

Component	Resource name	Role
Resource Group	25MCAR0141	Logical container for all experiment resources.
Storage Account	calablake141	Azure Data Lake Storage Gen2, rawdata container, student_performance.csv, 50 rows.
ML Workspace	calab-ml-141	Azure ML workspace, Central India, MLflow tracking enabled.
Data Asset	student-performance:1	Uniform resource identifier (URI) file data asset, student_performance.csv.
Compute Cluster	cpu-cluster	Azure ML compute, Standard_DS2_v2, minimum 0 nodes, maximum 2 nodes, idle scale down 120 seconds.
Sweep Job	stoic_date_sllzbmj5xh	Sweep job, experiment exp10-hyperdrive, 8 trials, random sampling, primary metric rmse, goal minimize.
Best Model	student-mathscore-rf:1	Registered model from the best trial, retrained with the best hyperparameters.

Source: Azure ML workspace, job, and model records, verified 2026-11-02.

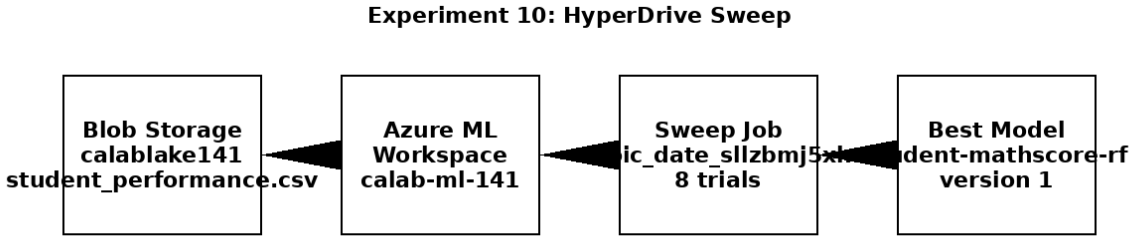


Figure 1.1 Experiment 10 flow

Blob Storage to Azure ML data asset to sweep job on cpu-cluster to registered best model.

4 Procedure

Step 1 (ML workspace). Create Azure ML workspace calab-ml-141 in resource group 25MCAR0141, Central India, with its managed storage account, key vault, and Application Insights. No dedicated pool or compute is created at this stage, so no hourly compute charge starts.

Step 2 (Data asset). Upload student_performance.csv as data asset student-performance version 1, type uri_file, in workspace calab-ml-141. The file has 50 data rows and 13 columns, the same file used in Experiments 07 to 09.

Step 3 (Compute cluster). Create compute cluster cpu-cluster, size Standard_DS2_v2, minimum 0 nodes, maximum 2 nodes, idle time before scale down 120 seconds. The DS2_v2 size was chosen after checking the regional quota for the DSv2 family, current 0, limit 4 virtual central processing units (vCPUs), in Central India. A DS2_v2 node uses 2 vCPUs, so 2 nodes fit exactly in the 4 vCPU quota, while a DS3_v2 node uses 4 vCPUs, so 2 DS3_v2 nodes would need 8 vCPUs and exceed the quota. The estimated cost was about Rs 30 to Rs 35 per hour while a node runs, and Rs 0 at 0 nodes, with the 8 trial sweep expected to take 10 to 15 minutes of cluster time.

Step 4 (Training script). Write train.py with a RandomForestRegressor, features Age, Gender, ReadingScore, WritingScore, AttendanceRate, and StudyHoursPerWeek, target MathScore, train test split 0.2 with random_state 42, median imputation for missing values, and MLflow logging of rmse, r2, and the three hyperparameters. The script takes n_estimators, max_depth, and min_samples_split as command line arguments, so the sweep job can set them per trial.

Step 5 (First sweep attempt, failed). Submit a sweep job with environment AzureML-sklearn-1.5-ubuntu20.04-py38-cpu. The submission failed with the error that no environment exists with that name in the workspace. The workspace environment list shows AzureML-sklearn-1.0-ubuntu20.04-py38-cpu, so the sweep definition was corrected to that environment, version 36, before resubmission. No trial ran in that first attempt.

Step 6 (Second sweep attempt, failed trials). Submit sweep job tidy_zoo_8hrlbtfgjq with the corrected environment. All trials failed with the error that the input contains not a number (NaN) values at model fit. The cause was that the first version of train.py dropped rows missing AttendanceRate or StudyHoursPerWeek only, but Age also has 5 missing values, which remained in the training data. The script was corrected to median imputation for all three columns, as described in Section 2, and the sweep was resubmitted. No metric from the failed job is used in this report.

Step 7 (Successful sweep). Submit sweep job stoic_date_sllzbmj5xh, display name exp10-hyperdrive-rf-mathscore, experiment exp10-hyperdrive, with search space n_estimators choice 10, 50, 100, max_depth choice 3, 5, 10, min_samples_split choice 2, 5, random sampling, maximum 8 trials, maximum 2 concurrent trials, timeout 3600 seconds, compute cpu-cluster, primary metric rmse, goal minimize. The job completed with 8 completed trials, 0 failed, and 0 canceled, in 12 minutes 7 seconds, as shown in Screenshot 2.

Step 8 (Best model retrain and registration). Retrain the best configuration as a separate command job teal_owl_f1rn3x3nk6, using train_best.py with n_estimators 100, max_depth 3, min_samples_split 2, the same split and imputation, saving model.pkl and metrics.json to outputs. Download model.pkl and register it as model student-mathscore-rf version 1 in workspace calab-ml-141, for deployment in Experiment 11. The retrained metrics match the best trial exactly, RMSE 1.487349 and R2 0.993999, which confirms that the retrain used the same data, split, and hyperparameters.

Step 9 (Compute cleanup, planned). Delete compute cluster cpu-cluster after the sweep and retrain finish, and verify that it is gone or at 0 nodes, so that no hourly compute charge continues. The cluster is set to scale to 0 nodes after 120 seconds idle, and deletion is the final step after Experiment 11, which reuses the registered model, not the cluster.

5 Proof of Work

The GitHub repository for this experiment is https://github.com/25MCAR0141/Cloud-Analytics-Azure/tree/main/Lab/Exp_10 . The repository contains the README, architecture diagram, Azure ML Studio screenshots, training scripts, sweep definition, best model retrain script, and the student performance dataset, without connection secrets. Table 1.2 records the sweep job, and Table 1.3 records all 8 trials in RMSE order, with the best trial first.

Table 1.2 Sweep job record, Experiment 10

Item	Value
Sweep job	stoic_date_sllzbmj5xh, display name exp10-hyperdrive-rf-mathscore, experiment exp10-hyperdrive
Status	Completed, 8 completed, 0 failed, 0 canceled, duration 12 minutes 7 seconds
Studio URL	https://ml.azure.com/runs/stoic_date_sllzbmj5xh?wsid=/subscriptions/cb9a06b3-8979-4a6c-a8fd-f924e54cd606/resourcegroups/25MCAR0141/workspaces/calab-ml-141

Item	Value
Primary metric	rmse, goal minimize, sampling RANDOM
Search space	n_estimators 10, 50, 100; max_depth 3, 5, 10; min_samples_split 2, 5
Compute	cpu-cluster, Standard_DS2_v2, minimum 0 nodes, maximum 2 nodes
Environment	AzureML-sklearn-1.0-ubuntu20.04-py38-cpu, version 36
Best trial	stoic_date_sllzbmj5xh_4, RMSE 1.487349, R2 0.993999
Best model job	teal_owl_f1rn3x3nk6, Completed, retrained best configuration, model.pkl saved
Registered model	student-mathscore-rf, version 1, type custom_model, workspace calab-ml-141

Source: Azure ML job and model records, and trial metrics.json files, verified 2026-11-02.

Table 1.3 Sweep trials, in RMSE order, best first

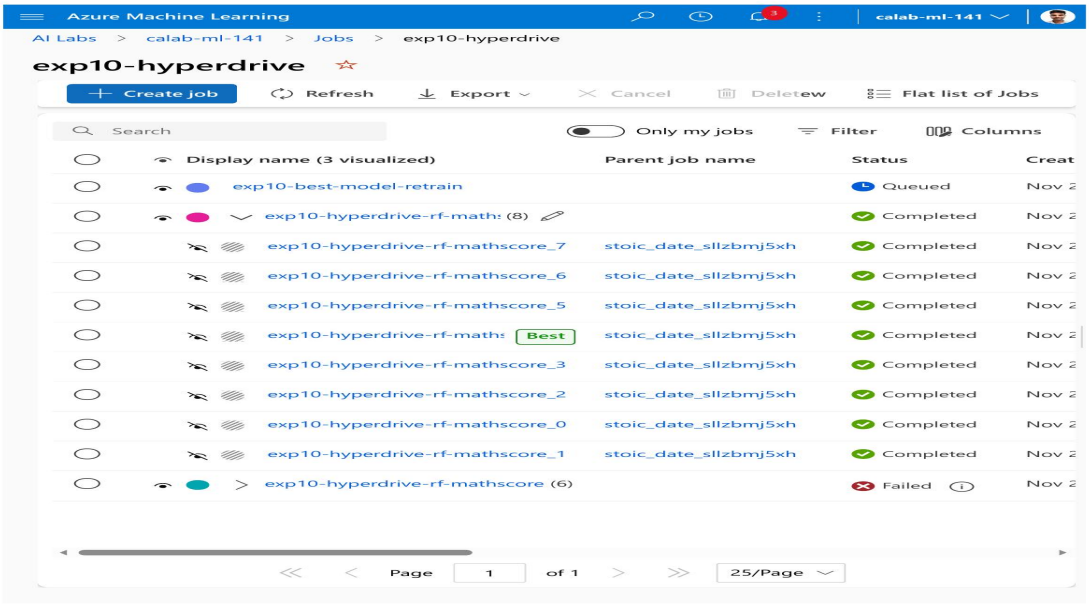
Trial	n_estimators	max_depth	min_samples_split	RMSE	R2
4	100	3	2	1.4873	0.9940
1	100	5	5	1.5294	0.9937
7	50	3	5	1.5625	0.9934
0	50	5	2	1.6033	0.9930
5	10	3	2	1.9369	0.9898
3	10	5	5	2.0644	0.9884
2	10	10	2	2.1891	0.9870
6	10	3	5	2.2515	0.9862

Source: Trial metrics.json files from jobs stoic_date_sllzbmj5xh_0 to _7, verified 2026-11-02. All trials used 50 rows, 40 training and 10 test. RMSE values are rounded to 4 decimal places in this table; full values are in the trial files and in Screenshot 3 for the best trial.

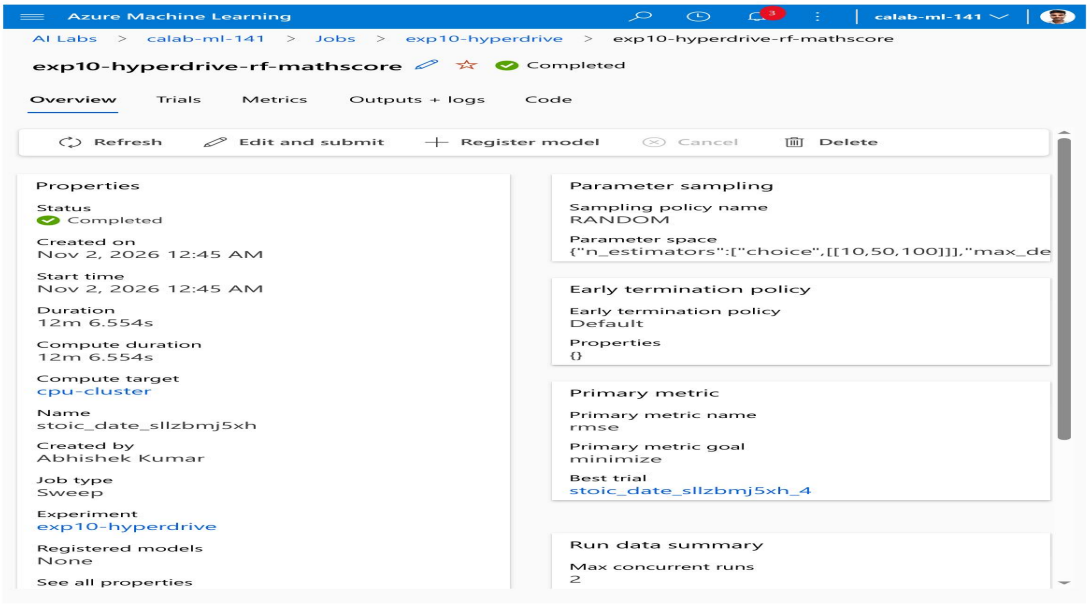
Result. The best configuration is 100 trees, maximum depth 3, and minimum samples split 2, with RMSE 1.4873 and R2 0.9940 on the 10 row test set. The pattern in Table 1.3 is that 100 tree and 50 tree configurations give lower RMSE than 10 tree configurations, and depth 3 gives the best result at 100 trees, which is consistent with a small dataset where a shallow forest generalizes better than a deep one. This result is specific to the 50 row test file and the 10 row test set, and it should not be generalized to larger student datasets without retraining.

6 Screenshots

The screenshots in this section are real Azure ML Studio page captures from the signed-in session on 2026-11-02, not replotted charts. The Studio capture shows the Studio viewport, not the operating system browser chrome, so the address bar and tabs are not drawn inside the images, as in Experiments 07 to 09. The Studio URLs are given in Section 5 and in each caption. Figure 1.1 in Section 3 remains the architecture diagram, and Tables 1.2 and 1.3 in Section 5 remain the job and trial records; this section is the visual proof of the Studio pages themselves.

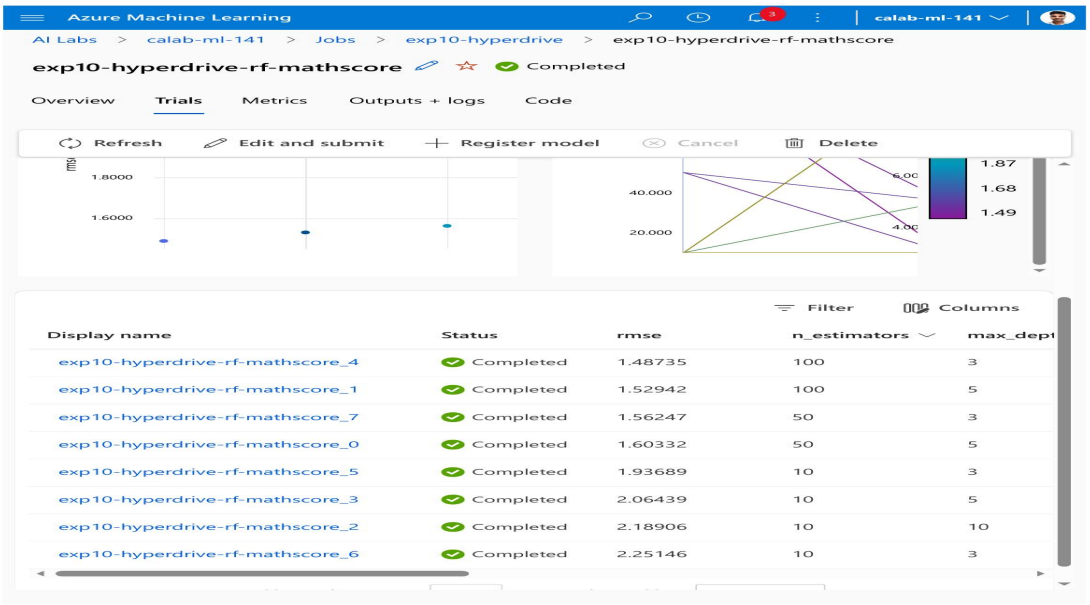


Screenshot 1 Azure ML Studio jobs list, experiment exp10-hyperdrive, sweep job expanded *Sweep exp10-hyperdrive-rf-mathscore (8), Completed, with 8 child trials Completed, trial _4 marked Best, parent job stoic_date_sllzbmj5xh.*



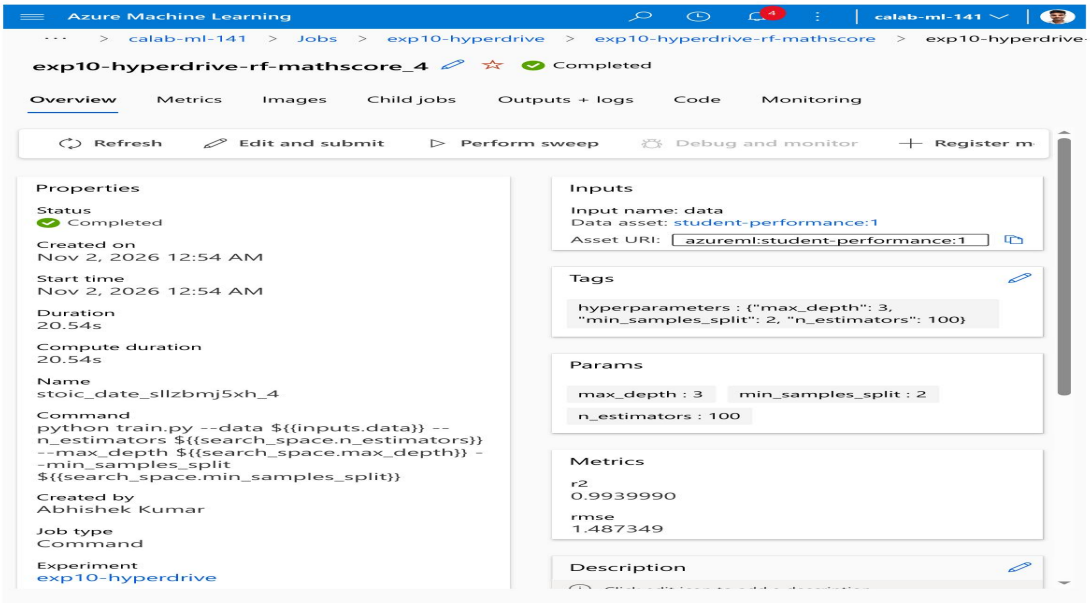
Screenshot 2 Azure ML Studio sweep job overview, stoic_date_sllzbmj5xh

Display name exp10-hyperdrive-rf-mathscore, Status Completed, Job type Sweep, Compute cpu-cluster, primary metric rmse, goal minimize, best trial stoic_date_sllzbmj5xh_4.



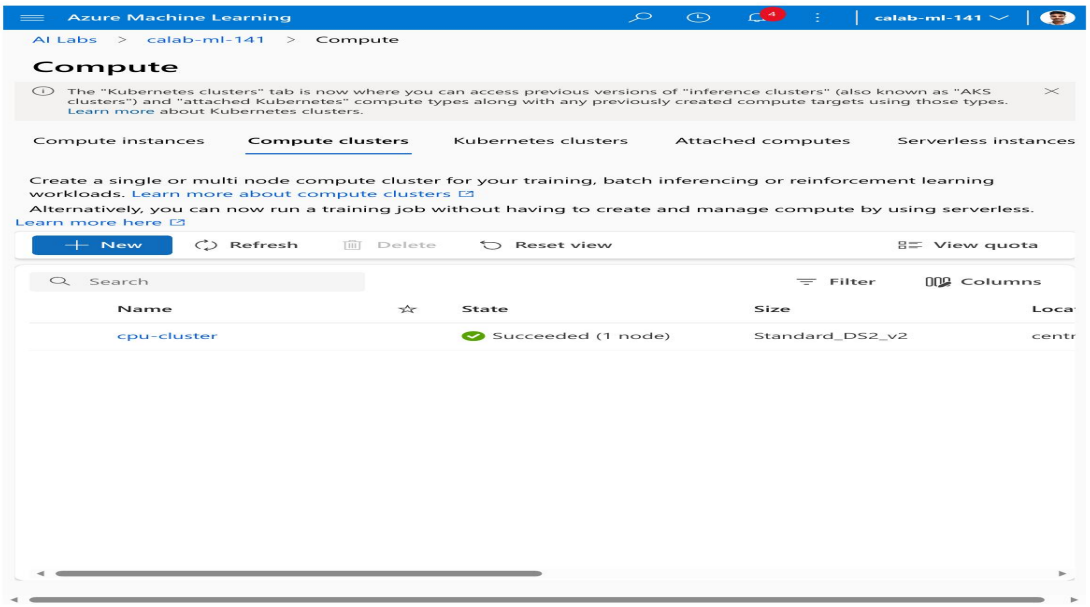
Screenshot 3 Azure ML Studio sweep trials list, stoic_date_sllzbmj5xh

Child runs 8 displayed, all Completed, with rmse values from 1.48735 for trial _4 to 2.25146 for trial _6.



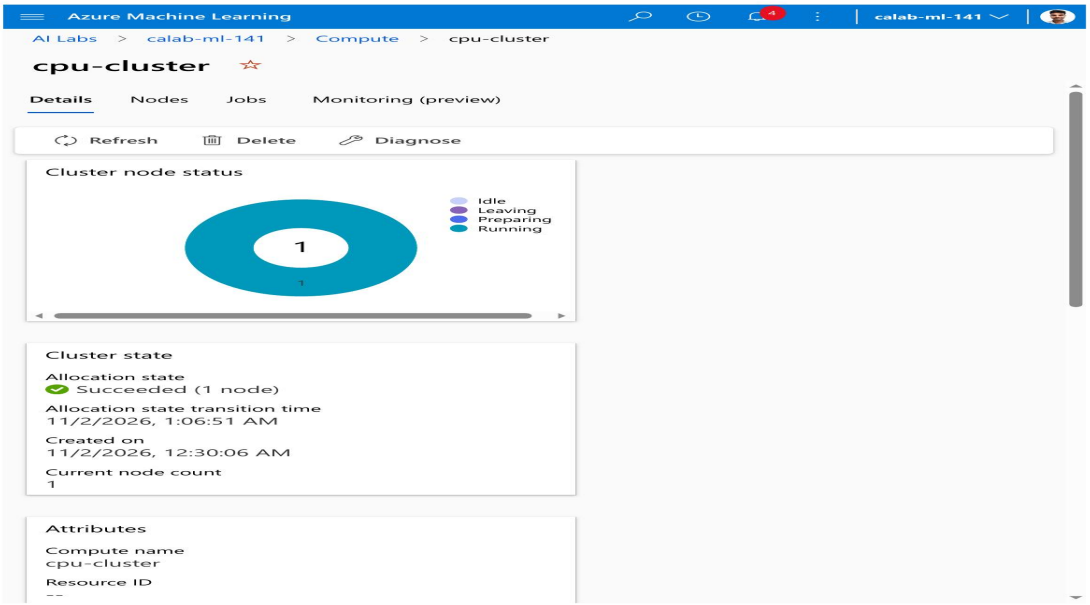
Screenshot 4 Azure ML Studio best trial overview, stoic_date_sllzbmj5xh_4

Status Completed, Duration 20.54 seconds, Compute cpu-cluster, metrics r2 0.9939990 and rmse 1.487349, params n_estimators 100, max_depth 3, min_samples_split 2.



Screenshot 5 Azure ML Studio compute clusters list

One compute shown: cpu-cluster, State Succeeded, Size Standard_DS2_v2, Location centralindia.



Screenshot 6 Azure ML Studio compute detail, cpu-cluster

Virtual machine size Standard_DS2_v2, 2 cores, 7 GB RAM, minimum nodes 0, maximum nodes 2, region centralindia, resource group 25MCAR0141.

Cost note. The only billable compute in this experiment is cpu-cluster, Standard_DS2_v2, at about Rs 30 to Rs 35 per hour while a node runs, and Rs 0 at 0 nodes, plus small storage and Application Insights charges. The sweep ran for 12 minutes 7 seconds, so the cluster cost for the sweep itself is a few rupees. Delete the cluster after Experiment 11, which reuses the registered model, to stop any further compute charge.

References

[1] Microsoft, “Tune hyperparameters for your model with Azure Machine Learning,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/how-to-tune-hyperparameters

[2] Microsoft, “Sweep jobs in Azure Machine Learning,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/how-to-create-sweep-jobs

[3] Microsoft, “Azure Machine Learning documentation,” Microsoft Learn. Available: https://learn.microsoft.com/en-us/azure/machine-learning/